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United States Patent	12398759
Kind Code	B2
Date of Patent	August 26, 2025
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Bearing unit with rolling bodies

Abstract

A bearing unit includes rolling bodies between radially inner and outer rings and a seal element extending between the rings, the seal element including a metallic shield and an elastomeric coating overmolded on the shield. The coating includes an anchor body mounted in a seat of the outer ring, and the anchor body includes a first anchor lip on an axially inner side of the anchor body contacting the seat of the radially outer ring and a second anchor lip located on an axially outer side of the anchor body contacting and forming a seal at a location on the radially outer ring. A concave radially outwardly facing connecting surface extends from the first anchor lip to the second anchor lip and is configured to reduce a compression of the anchor body when the anchor body is inserted into the seat.

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Appl. No.: 18/348609

Filed: July 07, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240019000 A1	Jan. 18, 2024

Foreign Application Priority Data

IT	102022000014854	Jul. 15, 2022
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Publication Classification

Int. Cl.: F16C33/78 (20060101); F16C19/06 (20060101)

U.S. Cl.:

CPC F16C33/783 (20130101); F16C19/06 (20130101)

Field of Classification Search

CPC: F16C (33/7826); F16C (33/783); F16C (33/784); F16C (33/7843); F16C (33/7853); F16C (33/7856)

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Background/Summary

CROSS-REFERENCE

(1) This application claims priority to Italian patent application no. 10 2022 000014854 filed on Jul. 15, 2023, the contents of which are fully incorporated herein by reference.

TECHNOLOGICAL FIELD

(2) The present disclosure is directed to a bearing unit with rolling bodies and having a seal element. Such a bearing unit is suitable in particular for use in the food and beverage industry.

BACKGROUND

(3) Bearing units having rolling elements or bodies which, as is known, are used to allow relative movement between one component or group and another component or group, are known. The bearing unit generally has a first component, for example a radially outer ring, and a second component, for example a radially inner ring. As is known, the radially inner ring is often rotatable

while the radially outer ring is stationary, whereas in other applications the outer ring rotates and the inner ring is stationary. In any case, in roller bearing units, the rotation of one ring with respect to the other is made possible by a plurality of rolling elements positioned between the cylindrical surface of one component and the cylindrical surface of the second component, which cylindrical surfaces are usually referred to as raceways. The rolling elements may be balls, cylindrical or conical rollers, needle rollers and similar rolling elements.

(4) It is also known practice for bearing units to have suitable seal elements (seals or sealing devices) to protect against external contaminants and to create a seal with respect to lubricant. Typically, the seal elements are made up of a shaped shield mounted by interference fit in suitable seats in the rings of the bearing unit, for example the radially outer ring. The shaped shield has an elastomer coating with one or more sealing lips and a protrusion which, in use, is inserted by interference fit, therefore deformably, in the seat of the radially outer ring. The sealing lips may, in use, make frictional contact with other components of the bearing unit, for example with the radially inner ring, which is rotatable. The sealing lips may also be in a “guttering” arrangement, in other words axially external and substantially parallel to one another, so as to create “labyrinth” seals, for example, with the casing containing the bearing unit. Such seal elements must be very efficient both from a functional point of view and in terms of reliability, throughout the entire life of the bearing unit.

(5) One of the drawbacks of known solutions is the procedure for assembling the seal element. To be specific, when the seal element is inserted in the seat in the radially outer ring, the elastomer protrusion is highly compressed and, not having the space necessary to adapt properly to the seat, gives rise to high tension forces that risk causing the seal element to come out of the seat in the outer ring. In particular, it is difficult to implement a seal element according to the prior art in the automatic bearing unit assembly process as solid anchoring of the seal element cannot be obtained.

SUMMARY

(6) It is an aspect of the present disclosure to produce a bearing unit with rolling bodies which does not have the above drawbacks.

(7) In particular, a bearing unit according to the present disclosure comprises a seal element having a new anchoring system for optimizing the automatic assembly process on the production line.

(8) An aspect of the present disclosure unit comprises a bearing unit having a radially outer ring that is stationary, a radially inner ring that is rotatable with respect to an axis of rotation (X), a row of rolling bodies interposed between the radially outer ring and the radially inner ring, and a seal element interposed between the radially inner ring and the radially outer ring. The seal element is stably anchored, in use, in a seat of the radially outer ring and comprises, in turn, a shaped metallic shield and an elastomer coating overmolded on the shaped shield and provided with a radially external anchor body. The anchor body includes a first radially external axially internal anchor lip, a second shaped shield and provided with a radially external anchor body. The anchor body includes a first radially external axially internal anchor lip and a radial external connecting surface. The first anchor lip is configured in use to be stably inserted in the seat of the radially external ring. The second anchor lip configured in use to improve the position in the seat of the seal element and the protection from external contamination, and the connecting surface is located between the first and second anchor lips and is configured to reduce the compression of the anchor body during the assembly process.

(9) Another aspect of the disclosure comprises a bearing unit that includes a radially outer ring configured to be fixed relative to a stationary support, a radially inner ring configured to rotate relative to the radially outer ring about an axis of rotation, a row of rolling bodies between the radially outer ring and the radially inner ring, and a seal element between the radially inner ring and the radially outer ring. The seal element includes a shaped metallic shield and an elastomeric coating overmolded on the shaped metallic shield, and the elastomeric coating includes an anchor body mounted in a seat of the radially outer ring. The anchor body is located radially outward of

the shaped metallic shield. The anchor body includes a first anchor lip on an axially inner side of the anchor body that contacts the seat of the radially outer ring, and a second anchor lip located on an axially outer side of the anchor body that contacts and forms a seal at a location on the radially outer ring. A concave, radially outwardly facing first connecting surface is located between the first anchor lip and the second anchor lip and is configured to reduce a compression of the anchor body when the anchor body is inserted into the seat.

(10) In another aspect of the disclosure, the first anchor lip comprises a first protrusion on the axially inner side of the anchor body and a second protrusion located between the first protrusion and the second anchor lip, and the first protrusion is connected to the second protrusion by a second connecting surface. The first protrusion contacts the seat at a first point, the second protrusion contacts the seat at a second point axially and radially outward of the first point, and a first sealed space is defined by the first protrusion, the second protrusion and a portion of the seat between the first point and the second point. Furthermore, the second anchor lip contacts the radially outer ring at a third point axially outward and radially inward of the second point, and a second sealed space is defined by the second protrusion, the second anchor lip and a portion of the radially outer bearing ring between the second point and the third point. A radially outer portion of the second anchor lip lies radially outward of the third point.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure will now be described with reference to the attached drawings, which show some non-limiting embodiments of the bearing unit, in which:
- (2) FIG. 1 is a sectional elevational view of a bearing unit with inner and outer rings, rolling bodies between the rings and a seal element including an anchor body according to an embodiment of the present disclosure.
- (3) FIG. 2 is an enlarged sectional view of one of the seal elements of FIG. 1.
- (4) FIG. 3 is a first schematic detail view of the anchor body of the seal element of FIG. 1.
- (5) FIG. 4 is a second schematic detail view of the anchor body of the seal element of FIG. 1.
- (6) FIG. 5 is a schematic detail view of an alternative embodiment of the anchor body of the seal element of FIG. 3.

DETAILED DESCRIPTION

- (7) An embodiment of a bearing unit with rolling bodies according to the present disclosure is described below purely by way of example.
- (8) With reference in particular to FIG. 1, the bearing unit **30** for use, preferably but not exclusively, in the food and beverage industry, comprises: a radially outer ring **31**, preferably stationary, a radially inner ring **33**, preferably rotatable about a central axis of rotation X of the bearing unit **30** and having at least one through hole **36** for receiving a device for attachment (known and therefore not shown in the figures, for example a pair of grub screws or plugs) of the bearing unit **30**, specifically the radially inner ring **33**, to a machine shaft, and a row of rolling elements **32** interposed between the radially outer ring **31** and the radially inner ring **33**.
- (9) Throughout the present description and in the claims, terms and expressions indicating positions and orientations, such as “radial” and “axial” are to be understood with reference to the central axis of rotation X of the bearing unit **30**. The term “axially inner” refers to a direction toward the rolling elements **32**, and the term “axially outer” refers to a direction away from the rolling elements **32**.
- (10) To simplify the drawings, the reference sign **32** will designate both individual rolling elements and the row of rolling elements. Again for the sake of simplicity, the term “rolling element” will be used to identify any rolling element, namely balls, rollers, conical rollers, needle rollers and the like, without thereby departing from the scope of the present invention.

(11) The bearing unit **30** also has seal elements **35** for sealing off the bearing unit from the external environment.

(12) The seal element **35** is interposed between the radially inner ring **33** and the radially outer ring **31**. In use, the seal element **35** is stably anchored in a seat **31a** in the radially outer ring **31**. The seat **31a** includes a groove that is located at the radially inner side of the radially outer ring **31** and has a toroidal surface.

(13) With reference to FIG. 2, the seal element **35** comprises a shaped metallic shield **40** that forms a support and an elastomer coating **50** overmolded on the shaped shield **40**.

(14) In particular, the shaped shield **40** comprises a first radially outer cylindrical portion **41** coaxial with the axis X, a second radially inner cylindrical portion **42**, again coaxial with the axis X, and a flange portion **43** connecting the first and second cylindrical portions, which flange extends radially inwards from said first cylindrical portion **41** and transversely to the axis X.

(15) The elastomer coating **50** comprises a contacting lip **51** that extends radially inward from the shaped shield **40** and slidably contacts the radially inner ring **33** to create a seal by frictional contact with the radially inner ring **33**. The elastomer coating **50** also includes at least one non-contacting lip **52** extending axially away from the shaped shield **40** and in a direction away from the rolling elements **32**. FIG. 2 shows an embodiment with a single non-contacting lip. However, solutions with several non-contacting lips, in a “guttering” arrangement, in other words axially external and substantially parallel to one another so as to create “labyrinth” seals with a casing (known and therefore not shown in the figures) containing the bearing unit **30** are also known and may be used, and an anchor body **55**, radially external of the shaped shield and rigidly secured to the first cylindrical portion **41** of the shaped shield **40**, making it possible to stably attach the seal element **35** to the seat **31a** in the radially outer ring **31**.

(16) The seal element includes an anchor body **55** that extends at least partially into the seat **31a** to secure the seal element **35** to the bearing radial outer ring **31**. The disclosed anchor body is configured to be more flexible than known elements for attaching a seal element to a bearing ring and to reduce tension forces linked to the deformation of the elastomeric material. The anchor body **55** has a shape which interfaces better with the seat **31a** in the radially outer ring **31** and which facilitates assembly in the seat **31a** in the radially outer ring **31**.

(17) The anchor body **55** includes a first anchor lip **60** that is located radially outward of the shaped shield **40** and on the axially inner side of the seal element **35**—that is, the side of the seal element **35** that faces toward the bearing interior where the rolling elements **32** are located. The first anchor lip **60** is configured in use to be stably inserted in the seat **31a** of the radially external ring **31**. The anchor body **55** also includes a second anchor lip **70** that is located radially outward of the shaped shield **40** and on the axially outer side of the seal element **35**—that is, on the side of the seal element that faces away from the bearing interior. The second anchor lip **70** is configured in use to improve the position in the seat **31a** of the seal element **35** and to form a seal with the bearing external ring **31** to protect the bearing from external contamination. More specifically, the second lip **70**, once placed in its operational anchoring position, becomes for all intents and purposes a static sealing lip on the radially outer ring **31**. It also creates a “dynamic seal” by interacting with a groove **53**, below the second lip **70**, and with the non-contacting lip **52**. This affords better protection against the ingress of contaminants both owing to the direct contact itself and owing to the curved shape of the groove **53** which tends to repel contaminants heading for the bearing unit **30**. The anchor body **55** also includes a convex connecting surface **80** or groove located radially outward of the shaped shield **40** between the first anchor lip **60** and the second anchor lip **70**. The connecting surface **80** is configured to reduce compression during the assembly process, thereby facilitating assembly of the seal element **35** in the seat **31a** in the radially outer ring **31**. According to a first embodiment of the disclosure, the connecting surface **80** is a toroidal surface. In an alternative embodiment, shown in FIG. 5, the connecting surface **81** is a cylindrical surface.

(18) More specifically, the first anchor lip **60** has an axially inwardly facing first protrusion **61**, and

a second protrusion **62** located axially and radially outward from the first protrusion **61**. A slightly concave conical surface **63** connects the first protrusion **61** to the second protrusion **62** and is configured to reduce the tension on the anchor body **55** during assembly of the seal element **35**.

(19) The seat **31a** in the radially outer ring **31** has a conical surface **311** located axially inward of the shaped shield **40**, a circular surface **312** axially outward of the conical surface **311**, and a substantially toroidal surface **313** between the conical surface **311** and the circular surface **312**. The radially outer ring **31** also has an edge **31b** axially external and radially internal to the seat **31a**.

(20) The seal element **35** is assembled in the seat **31a** by inserting, in the axially internal direction, the first anchor lip **60**, which deforms (specifically, its second protrusion **62** deforms in particular) so as to slide against and pass beyond the edge **31b** of the radially outer ring **31**. Next, the first anchor lip **60** enters the seat **31a**, relaxing and taking up a position, shown schematically in FIG. 3, affording stable anchoring by virtue of the reaction forces exchanged, respectively, between the first protrusion **61** and the conical surface **311** of the seat **31a** and between the second protrusion **62** and the circular surface **312** of the seat **31a**.

(21) The double-lip design of the anchor body **55** has two simultaneous functions: the first axially internal anchor lip **60** enters the seat **31a** in the radially outer ring **31** “gently” and stays rigidly attached to said seat; the second anchor lip **70**, which is external, is basically a second protection both for stabilizing the assembly process through the contact with a radially internal conical surface **31b'** of the edge **31b** of the radially outer ring **31**, and for creating a contacting seal with the outer ring against external contaminants.

(22) Therefore, this new shape of the anchor body **55**, in particular of the first anchor lip **60**, stabilizes the position of the seal element **35** and the forces within the seat **31a** in the radially outer ring **31**. Furthermore, also by virtue of the conical surface **63**, the first anchor lip **60**, and therefore the entire anchor body **55**, is not subjected to high tension forces that could cause it to come out of the corresponding seat **31a**.

(23) It will be understood that, in order to ensure the highest efficiency of this new anchor body **55**, it is advantageous to combine this new solution with optimized dimensions.

(24) With reference to FIG. 4, as h_1 defines the axial width of the seat **31a** and in particular of its toroidal central surface **313**, the fundamental dimensions of the anchor body **55** may be related directly or indirectly thereto.

(25) The axial width A of the first anchor lip **60**, in other words the axial distance from the axially inner edge **611** of the first protrusion **61** to the axially outer edge **621** of the second protrusion **62**, may be between

$$h_1 < A < 1.1 \times h_1,$$

wherein higher values would excessively stress the anchor body **55**, while lower values would not guarantee sufficient stability of the anchor body in the seat.

(26) The axial width D of the entire anchor body **55**, in other words the axial distance from the axially inner edge **611** of the first protrusion **61** to the axially outer edge **701** of the second anchor lip **70**, may be between

$$2.0 \times A < D < 2.1 \times A$$

wherein higher values would not guarantee a sufficient seal between the second anchor lip **70** and the conical surface **31b'** of the edge **31b** of the radially outer ring **31**, while lower values would excessively stress the second anchor lip **70**.

(27) The radial width C of the second anchor lip **70**, that is the radial distance between the radially outer surface **411** of the first cylindrical portion **41** of the shaped shield **40** and a cylindrical surface **622**, radially external, of the second protrusion **62**, may be between

$$1.2 \times A < C < 1.25 \times A.$$

(28) In practice, this makes it possible to restrict as far as possible the value of the thickness of elastomer of the first anchor lip **60** and, therefore, the radial height of the entire seal element for better insertion in the seat in the radially outer ring. Higher values than the selected range would

make assembly of the seal element more difficult, while lower values would not guarantee optimum production of the two anchor lips **60**, **70** for reasons of the production process.

(29) The radius of curvature R1 of the slightly concave conical surface **63** may be between $1.4 \times A < R1 < 1.45 \times A$

in order to guarantee proper flexibility of the first anchor lip **60**.

(30) The radius of curvature R2 of the radially outer connecting surface **80** between the first anchor lip **60** and the second anchor lip **70** may be between 0.15 mm and 1 mm in order to guarantee proper compression of the first anchor lip **60** towards the connecting surface **80**, **81** in such a way as to facilitate insertion of the profile of the anchor body **55** in the seat **31a**, during the assembly process.

(31) Lastly, the radius R3 of the axially inner edge **611** of the first protrusion **61** may be between 0.15 mm and 1.5 mm, in order to guarantee correct positioning of the entire seal element **35** in the seat **31a** by compressing the two protrusions **61**, **62** in such a way as to balance the reaction forces exchanged, respectively, between the first protrusion **61** and the conical surface **311** of the seat **31a** and between the second protrusion **62** and the circular surface **312** of the seat **31a**.

(32) In summary, the solution proposed by the seal element with an optimized anchor body offers the following advantages: improves automatic assembly of the seal element in the bearing unit, preventing it from coming out during application, and also preventing the elastomeric material from being squashed on the edge of the outer ring with consequent tearing of the seal element in the assembly phase, generally improves the process for assembling the seal element, preventing it from rotating relative to the radially outer ring, does not change the current process for assembling the bearing unit, introducing further assembly steps, does not change the seal element as regards the sealing lips, since the changes proposed only affect the anchoring area.

(33) In addition to the embodiments of the invention as described above, it is to be understood that there are numerous other variants. It is also to be understood that said embodiments are solely exemplary and do not limit the scope of the invention, its applications, or its possible configurations. On the contrary, although the above description enables those skilled in the art to apply the present invention in at least one exemplary configuration, it is to be understood that numerous variations of the described components may be devised, without thereby departing from the scope of the invention as defined in the appended claims, interpreted literally and/or according to their legal equivalents.

(34) Representative, non-limiting examples of the present invention were described above in detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing preferred aspects of the present teachings and is not intended to limit the scope of the invention. Furthermore, each of the additional features and teachings disclosed above may be utilized separately or in conjunction with other features and teachings to provide improved bearing units.

(35) Moreover, combinations of features and steps disclosed in the above detailed description may not be necessary to practice the invention in the broadest sense, and are instead taught merely to particularly describe representative examples of the invention. Furthermore, various features of the above-described representative examples, as well as the various independent and dependent claims below, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.

(36) All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter.

Claims

1. A bearing unit comprising: a radially outer ring configured to be fixed relative to a stationary support, a radially inner ring configured to rotate relative to the radially outer ring about an axis of rotation, a row of rolling bodies between the radially outer ring and the radially inner ring, a seal element between the radially inner ring and the radially outer ring, the seal element including a shaped metallic shield and an elastomeric coating overmolded on the shaped metallic shield, wherein the elastomeric coating includes an anchor body mounted in a seat of the radially outer ring, the anchor body being located radially outward of the shaped metallic shield, wherein the anchor body includes a first anchor lip on an axially inner side of the anchor body, the first anchor lip contacting the seat of the radially outer ring, a second anchor lip located on an axially outer side of the anchor body, the second anchor lip contacting and forming a seal at a location on the radially outer ring, and a radially outwardly facing first connecting surface between the first anchor lip and the second anchor lip configured to reduce a compression of the anchor body when the anchor body is inserted into the seat, the first connecting surface being concave, wherein the first anchor lip comprises a first protrusion on the axially inner side of the anchor body and a second protrusion located between the first protrusion and the second anchor lip, and wherein the first protrusion is connected to the second protrusion by a second connecting surface.
2. The bearing unit according to claim 1, wherein the second connecting surface is concave.
3. The bearing unit according to claim 2, wherein a radius of curvature of the first connecting surface is less than a radius of curvature of the second connecting surface.
4. The bearing unit according to claim 2, wherein an axial distance A from an axially inner edge of the first protrusion to an axially outer edge of the second protrusion satisfies an inequality $h1 < A < 1.1 \times h1$, where $h1$ is an axial width of the seat.
5. The bearing unit according to claim 4, wherein an axial distance D from the axially inner edge of the first protrusion to an axially outer edge of the second anchor lip satisfies an inequality: $2.0 \times A < D < 2.1 \times A$.
6. The bearing unit according to claim 4, wherein a radial distance C between a radially outer surface of a first cylindrical portion of the shaped metallic shield and a cylindrical surface of the second protrusion satisfies an inequality: $1.2 \times A < C < 1.25 \times A$.
7. The bearing unit according to claim 4, wherein a radius of curvature $R1$ of the first connecting surface satisfies an inequality: $1.4 \times A < R1 < 1.45 \times A$.
8. The bearing unit according to claim 4, wherein a radius ($R3$) of the axially inner edge of the first protrusion is between 0.15 mm and 1.5 mm.
9. The bearing unit according to claim 4, wherein an axial distance D from the axially inner edge of the first protrusion to an axially outer edge of the second anchor lip satisfies an inequality: $2.0 \times A < D < 2.1 \times A$, wherein a radial distance C between a radially outer surface of a first cylindrical portion of the shaped metallic shield and a cylindrical surface of the second protrusion satisfies an inequality: $1.2 \times A < C < 1.25 \times A$, wherein a radius of curvature $R1$ of the concave surface satisfies an inequality: $1.4 \times A < R1 < 1.45 \times A$, wherein a radius of curvature $R2$ of the second connecting surface is between 0.15 mm and 1 mm, and wherein a radius ($R3$) of the axially inner edge of the first protrusion is between 0.15 mm and 1.5 mm.
10. The bearing unit according to claim 2, wherein the first protrusion contacts the seat at a first point, wherein the second protrusion contacts the seat at a second point axially and radially outward of the first point, wherein a first sealed space is defined by the first protrusion, the second protrusion and a portion of the seat between the first point and the second point.

11. The bearing unit according to claim 10, wherein the second anchor lip contacts the radially outer ring at a third point axially outward and radially inward of the second point, and wherein a second sealed space is defined by the second protrusion, the second anchor lip and a portion of the radially outer bearing ring between the second point and the third point.
 12. The bearing unit according to claim 11, wherein a radially outer portion of the second anchor lip lies radially outward of the third point.
 13. The bearing unit according to claim 1, wherein the second anchor lip sealingly contacts a radially inner surface of the radially outer ring.
 14. The bearing unit according to claim 13, wherein a radius of curvature R2 of the second connecting surface is between 0.15 mm and 1 mm.
 15. The bearing unit according to claim 1, wherein the first connecting surface is toroidal or cylindrical.
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